

A Priori Algorithm

Association Rule Mining

- Association rule mining is a popular, unsupervised learning technique, used in business to help identify shopping patterns.
- It is also known as market basket analysis.
- It helps find interesting relationships (affinities) between variables (items or events).
- Thus, it can help cross-sell related items and increase the size of a sale.

The Market-Basket Model

- A large set of **items**
 - e.g., things sold in a supermarket
- A **large set** of **baskets**
- Each basket is a **small subset of items**
 - e.g., the things one customer buys on one day
- Want to discover **association rules**
 - People who bought $\{x,y,z\}$ tend to buy $\{v,w\}$
 - Amazon!

Input:

<i>TID</i>	<i>Items</i>
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Output:

Rules Discovered:

$\{\text{Milk}\} \rightarrow \{\text{Coke}\}$

$\{\text{Diaper, Milk}\} \rightarrow \{\text{Beer}\}$

Business Applications of Association Rules

- In business environments a pattern or knowledge can be used for many purposes.
- In sales and marketing, it is used for e-commerce site design, online advertising optimization, product pricing, and sales/promotion configurations.
- This analysis can suggest not to put one item on sale at a time, and instead to create a bundle of products promoted as a package to sell other non-selling items.
- In retail environments, it can be used for store design.
- Strongly associated items can be kept close together for customer convenience.
- Or they could be placed far from each other so that the customer has to walk the aisles and by doing so is potentially exposed to other items.

Frequent Itemsets

- **Simplest question:** Find sets of items that appear together “frequently” in baskets
- **Support** for itemset I : Number of baskets containing all items in I
 - (Often expressed as a fraction of the total number of baskets)
- Given a **support threshold s** , then sets of items that appear in at least s baskets are called **frequent itemsets**

<i>TID</i>	<i>Items</i>
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Support of
 $\{\text{Beer, Bread}\} = 2$

Example: Frequent Itemsets

- **Items** = {milk, coke, pepsi, beer, juice}

- **Support threshold** = 3 baskets

$$B_1 = \{m, c, b\}$$

$$B_2 = \{m, p, j\}$$

$$B_3 = \{m, b\}$$

$$B_4 = \{c, j\}$$

$$B_5 = \{m, p, b\}$$

$$B_6 = \{m, c, b, j\}$$

$$B_7 = \{c, b, j\}$$

$$B_8 = \{b, c\}$$

- **Frequent itemsets:** {m}, {c}, {b}, {j}, {m,b}, {b,c}, {c,j}

Association Rules

- **Association Rules:**

If-then rules about the contents of baskets

- $\{i_1, i_2, \dots, i_k\} \rightarrow j$ means: “if a basket contains all of $i_1, \dots, i_k$ then it is *likely* to contain j ”

- In practice there are many rules, want to find significant/interesting ones!

- **Confidence** of this association rule is the probability of j given $I = \{i_1, \dots, i_k\}$

$$\text{conf}(I \rightarrow j) = \frac{\text{support}(I \cup j)}{\text{support}(I)}$$

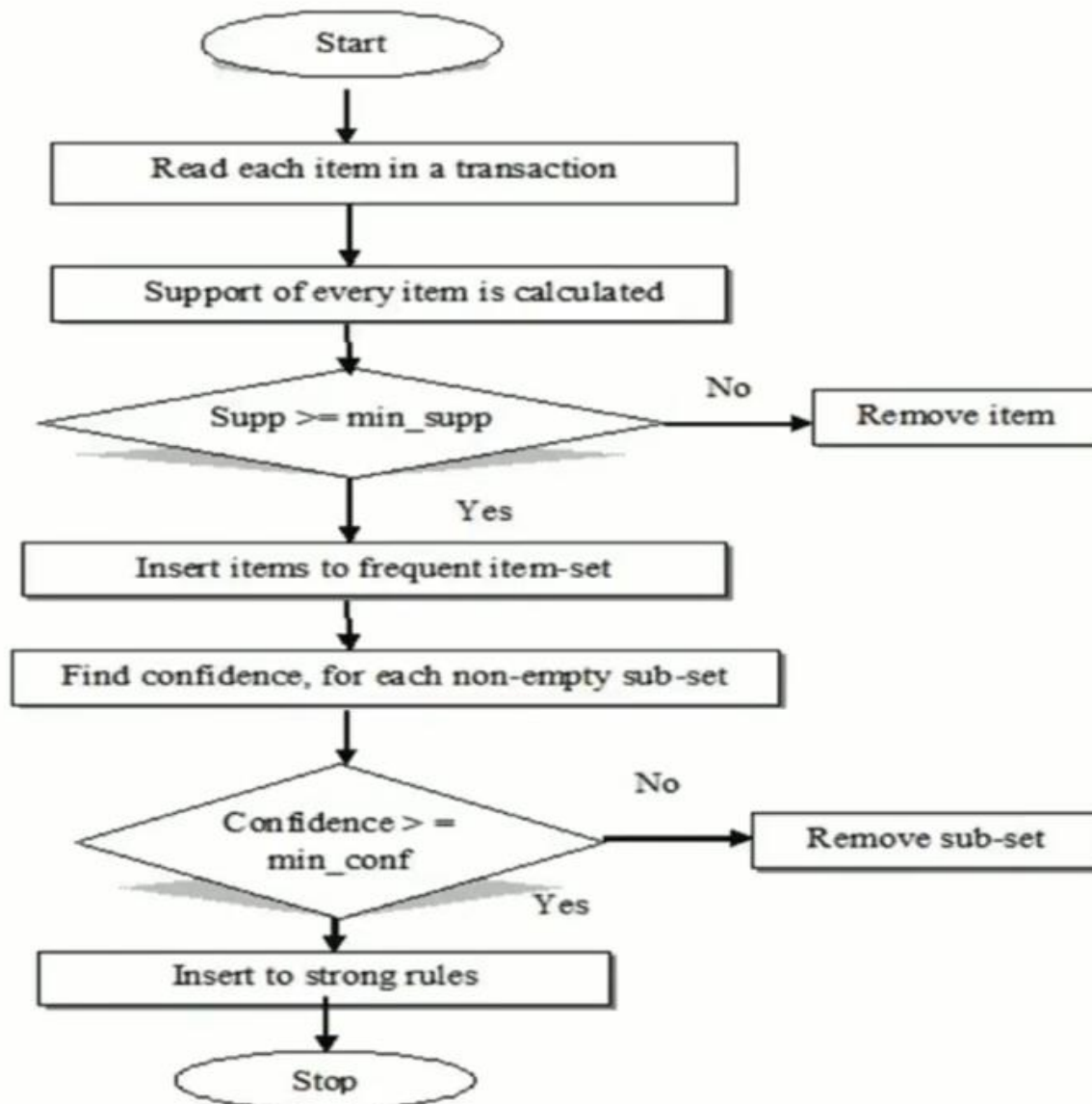
Algorithms for Association Rule

- Not all association rules are interesting and useful, only those that are strong rules and also those that occur frequently.
- In association rule mining, the goal is to find all rules that satisfy the user-specified ***minimum support*** and ***minimum confidence***.
- The resulting sets of rules are all the same irrespective of the algorithm used, that is, given a transaction data set T , a minimum support and a minimum confidence, the set of association rules existing in T is *uniquely determined*.
- The most popular algorithms are Apriori, Eclat, and FP-growth, along with various derivatives and hybrids of the three.

Apriori Algorithm

- This is the most popular algorithm used for association rule mining.
- A frequent itemset is an itemset whose support is greater than or equal to minimum support threshold.
- The Apriori property is a downward closure property, which means that any subsets of a frequent itemset are also frequent itemsets.
- Thus, if (A,B,C,D) is a frequent itemset, then any subset such as (A,B,C) or (B,D) are also frequent itemsets.
- This uses a bottom-up approach; and the size of frequent subsets is gradually increased, from one-item subsets to two-item subsets, then three-item subsets and so on.
- Groups of candidates at each level are tested against the data for minimum support.

Apriori Algorithm



Example - 1

Image ID	Associated Tags
1	Beach, Sunshine, Holiday
2	Sand, Beach
3	Sunshine, Beach, Ocean
4	Ocean, People, Beach, Sunshine
5	Holiday, Sunshine

- For the given dataset, apply Apriori algorithm to discover strong association rules among image tags.
- Assume that min_support=40% and min_confidence=70%.
- Generate association rules from the frequent itemsets. Calculate the confidence of each rule and identify all the strong association rules.

Example - 1

Image ID	Associated Tags
1	<u>Beach</u> , Sunshine, Holiday ✓
2	Sand, <u>Beach</u>
3	Sunshine, <u>Beach</u> , Ocean
4	Ocean, People, <u>Beach</u> , Sunshine
5	Holiday, ✓ <u>Sunshine</u>

1-Itemset	Support_count
Beach	4
Holiday	2
Ocean	2
People	1
Sand	1
Sunshine	4

min_support = 40%, ✓

min_support_count

= min_support x itemset_count

= 40% x 5

= 2

1-Frequent Itemset	Support_count
<u>Beach</u>	4
Holiday	2
Ocean	2
Sunshine	4

Example - 1

Image ID	Associated Tags
1	Beach, Sunshine, Holiday
2	Sand, Beach
3	Sunshine, Beach, Ocean
4	Ocean, People, Beach, Sunshine
5	Holiday, Sunshine

2-Itemset	Support_count
Beach, Holiday	1
Beach, Ocean	2
Beach, Sunshine	3
Holiday, Ocean	0
Holiday, Sunshine	2
Ocean, Sunshine	2

Beach, Holiday,
Ocean, Sunshine

2-Frequent Itemset	Support_count
Beach, Ocean	2
Beach, Sunshine	3
Holiday, Sunshine	2
Ocean, Sunshine	2

Example - 1

Image ID	Associated Tags
1	Beach, Sunshine, Holiday
2	Sand, Beach
3	Sunshine, Beach, Ocean
4	Ocean, People, Beach, Sunshine
5	Holiday, Sunshine

Beach, Ocean, Holiday, Sunshine

3-Itemset	Support_count
Beach, Ocean, Holiday	0
Beach, Ocean, Sunshine	2 ✓
Beach, Holiday, Sunshine	1
Ocean, Holiday, Sunshine	0

3-Frequent Itemset	Support_count
Beach, Ocean, Sunshine	2

Generate association rules from the frequent item sets. Calculate the confidence of each rule and identify all the strong association rules.

Example - 1

1-Frequent Itemset	Support_count
Beach	4
Holiday	2
Ocean	2
Sunshine	4

2-Frequent Itemset	Support_count
Beach, Ocean	2
Beach, Sunshine	3
Holiday, Sunshine	2
Ocean, Sunshine	2

3-Frequent Itemset	Support_count
Beach, Ocean, Sunshine	2

- **min_confidence = 70%** ✓
- Confidence $(X \rightarrow Y) = P(Y | X) = \frac{P(X \cup Y)}{P(X)}$
- We have 5 frequent itemsets:
- {Beach, Ocean}, {Beach, Sunshine}, {Holiday, Sunshine}, {Ocean, Sunshine} and {Beach, Ocean, Sunshine}.
- Therefore, candidate rules are:
- For {Beach, Ocean},
 - Beach > Ocean = $\frac{2}{4} = 50\%$
 - Ocean > Beach = $\frac{2}{2} = 100\%$ (Strong) ✓
- For {Beach, Sunshine},
 - Beach > Sunshine = $\frac{3}{4} = 75\%$ (Strong) ✓
 - Sunshine > Beach = $\frac{3}{4} = 75\%$ (Strong)

Example - 1

1-Frequent Itemset	Support_count
Beach	4
Holiday	2
Ocean	2
Sunshine	4

2-Frequent Itemset	Support_count
Beach, Ocean	2
Beach, Sunshine	3
Holiday, Sunshine	2
Ocean, Sunshine	2

3-Frequent Itemset	Support_count
Beach, Ocean, Sunshine	2

- **min_confidence = 70%**
- Confidence $(X \rightarrow Y) = P(Y | X) = P(X \cup Y) / P(X)$
- We have 5 frequent itemsets:
 - {Beach, Ocean}, {Beach, Sunshine}, {Holiday, Sunshine}, {Ocean, Sunshine} and {Beach, Ocean, Sunshine}.
- Therefore, candidate rules are:
- For {Holiday, Sunshine},
 - Holiday > Sunshine = $2/2 = 100\%$ (Strong) ✓
 - Sunshine > Holiday = $2/4 = 50\%$ ✗
- For {Ocean, Sunshine},
 - Ocean > Sunshine = $2/2 = 100\%$ (Strong) ✓
 - Sunshine > Ocean = $2/4 = 50\%$ ✗

Example - 1

1-Frequent Itemset	Support_count
Beach	4
Holiday	2
Ocean	2
Sunshine	4

2-Frequent Itemset	Support_count
Beach, Ocean	2
Beach, Sunshine	3
Holiday, Sunshine	2
Ocean, Sunshine	2

3-Frequent Itemset	Support_count
Beach, Ocean, Sunshine	2

- **min_confidence = 70%**
- Confidence $(X \rightarrow Y) = P(Y | X) = P(X \cup Y) / P(X)$
- We have 5 frequent itemsets:
- {Beach, Ocean}, {Beach, Sunshine}, {Holiday, Sunshine}, {Ocean, Sunshine} and {Beach, Ocean, Sunshine}.
- Therefore, candidate rules are:
- For {Beach, Ocean, Sunshine},
 - Beach > Ocean, Sunshine = $2/4 = 50\%$ ✗ ✓
 - Ocean, Sunshine > Beach = $2/2 = 100\%$ (Strong)
 - Ocean > Beach, Sunshine = $2/2 = 100\%$ (Strong)
 - Beach, Sunshine > Ocean = $2/3 = \sim 67\%$ ✗
 - Sunshine > Beach, Ocean = $2/4 = 50\%$ ✗
 - Beach, Ocean > Sunshine = $2/2 = 100\%$ (Strong)

Example - 2

T ID	Items Bought
1	{Bread, Butter, Milk}
2	{Bread, Butter}
3	{Beer, Cookies, Diapers}
4	{Milk, Diapers, Bread, Butter}
5	{Beer, Diapers}

- For the given dataset, apply Apriori algorithm to discover strong association rules among image tags.
- Assume that min_support=40% and min_confidence=70%.
- Generate association rules from the frequent itemsets. Calculate the confidence of each rule and identify all the strong association rules.

Example - 2

T ID	Items Bought
1	{Bread, Butter, Milk}
2	{Bread, Butter}
3	{Beer, Cookies, Diapers}
4	{Milk, Diapers, Bread, Butter}
5	{Beer, Diapers}

Generate candidate itemsets (C_k) and qualified frequent itemsets (L_k) step by step until the largest frequent itemset is generated.

Example - 2

T ID	Items Bought
1	{Bread, Butter, Milk}
2	{Bread, Butter}
3	{Beer, Cookies, Diapers}
4	{Milk, Diapers, Bread, Butter}
5	{Beer, Diapers}

Generate candidate itemsets (C_k) and qualified frequent itemsets (L_k) step by step until the largest frequent itemset is generated.

Example - 2

T ID	Items Bought
1	{ <u>B</u> read, But <u>t</u> er, Milk}
2	{ <u>B</u> read, But <u>t</u> er}
3	{Beer, Cookies, Diapers}
4	{Milk, Diapers, Bread, But <u>t</u> er}
5	{Beer, Diapers}

1-Itemset	Support_count
Bread	3
Butter	3
Milk	2
Beer	2
Cookies	1
Diapers	3

Bread, Butter, Milk,
Diapers, Beer

$\text{min_support} = \underline{40\%}, \checkmark$
 min_support_count
 $= \text{min_support} \times \text{itemset_count}$
 $= 40\% \times 5$
 $= \underline{2}$

1-Frequent Itemset	Support_count
Bread	3
Butter	3
Diapers	3
Milk	2
Beer	2

Example - 2

T ID	Items Bought
1	{Bread, Butter, Milk}
2	{Bread, Butter}
3	{Beer, Cookies, Diapers}
4	{Milk, Diapers, Bread, Butter}
5	{Beer, Diapers}

2-Itemset	Support_count
Bread, Butter	3 ✓
Bread, Diapers	1
Bread, Milk	2
Bread, Beer	0
Butter, Diapers	1
Butter, Milk	2
Butter, Beer	0
Diapers, Milk	1
Diapers, Beer	2
Milk, Beer	0

**Bread, Butter, Diapers,
Milk, Beer**

2-Frequent Itemset	Support_count
Bread, Butter	3
Bread, Milk	2
Butter, Milk	2
Diapers, Beer	2

Example - 2

T ID	Items Bought
1	<u>{Bread, Butter, Milk}</u>
2	{Bread, Butter}
3	{Beer, Cookies, Diapers}
4	{Milk, Diapers, Bread, Butter}
5	{Beer, Diapers}

Bread, Butter, Milk, Diapers, Beer

3-Itemset	Support_count
Bread, Butter, Milk	2 ✓
Bread, Butter, Diapers	1 ✓
Bread, Butter, Beer	0
Bread, Milk, Diapers	1
Bread, Milk, Beer	0
Bread, Diapers, Beer	0
Butter, Milk, Diapers	1
Butter, Milk, Beer	0
Butter, Diapers, Beer	0
Milk, Diapers, Beer	0
3-Frequent Itemset	Support_count
Bread, Butter, Milk	2

Example - 2

1-Frequent Itemset	Support_count
Bread	3
Butter	3
Diapers	3
Milk	2
Beer	2

2-Frequent Itemset	Support_count
Bread, Butter	3
Bread, Milk	2
Butter, Milk	2
Diapers, Beer	2

3-Frequent Itemset	Support_count
Bread, Butter, Milk	2

- **min_confidence = 70%**
- Confidence $(X \rightarrow Y) = P(Y | X) = P(X \cup Y) / P(X)$
- We have 5 frequent itemsets:
- {Bread, Butter}, {Bread, Milk}, {Butter, Milk}, {Diapers, Beer} and {Bread, Butter, Milk}.
- Therefore, candidate rules are:
- For {Butter, Milk}
 - butter>milk = $2/3 = 67\%$ ✗
 - milk>butter = $2/2 = 100\%$ (Strong) ✓
- For {Diapers, Beer}
 - diapers>beer = $2/3 = 67\%$ ✗
 - beer>diapers = $2/2 = 100\%$ (Strong)

Example - 2

1-Frequent Itemset	Support_count
Bread	3
Butter	3
Diapers	3
Milk	2
Beer	2

2-Frequent Itemset	Support_count
Bread, Butter	3
Bread, Milk	2
Butter, Milk	2
Diapers, Beer	2

3-Frequent Itemset	Support_count
Bread, Butter, Milk	2

- **min_confidence = 70%**
- Confidence $(X \rightarrow Y) = P(Y | X) = P(X \cup Y) / P(X)$
- We have 5 frequent itemsets:
- **{Bread, Butter}, {Bread, Milk}, {Butter, Milk}, {Diapers, Beer} and {Bread, Butter, Milk}.**
- Therefore, candidate rules are:
- **For {Bread, Butter, Milk}**
 - bread,butter>milk = $2/3 = 67\%$ ✗
 - bread,milk>butter = $2/2 = 100\%$ (Strong) ✓
 - milk,butter>bread = $2/2 = 100\%$ (Strong) ✓
 - bread>butter,milk = $2/3 = 67\%$ ✗
 - butter>bread,milk = $2/3 = 67\%$ ✗
 - milk>bread,butter = $2/2 = 100\%$ (Strong)